

José Luis Carmona Jiménez

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Employment

Postdoctoral Researcher 16/09/2024–30/06/2026

Simion Stoilow Institute of Mathematics of the Romanian Academy (IMAR), Bucharest, Romania.

Postdoctoral position supported by the PNRR-III-C9-2023-I8 grant CF 149/31.07.2023, *Conformal Aspects of Geometry and Dynamics*.

Secondary and Upper-Secondary School Teacher 01/09/2022–10/09/2024

Spain.

Permanent position.

Predoctoral Researcher in Differential Geometry 01/10/2020–31/08/2022

Universidad Complutense de Madrid (UCM), Madrid, Spain.

FPI predoctoral fellow in the research group *Geometrical, Topological and Combinatorial Structures Associated with Dynamical Systems*. Principal Investigator: Prof. Marco Castrillón López.

Researcher in Differential Geometry 01/07/2020–30/09/2020

Instituto de Ciencias Matemáticas (ICMAT), Madrid, Spain.

Research grant associated with the research group *Algebra and Geometry*.

Education

Ph.D. in Mathematics 2024

Universidad Complutense de Madrid, Spain

Thesis: *Homogeneous Descriptions and Families of Homogeneous Structures*. Advisor: Prof. Marco Castrillón López. Grade: *Sobresaliente Cum Laude*; International Mention.

M.A. in Secondary Mathematics Education 2019

University of Granada, Spain.

M.Sc. in Advanced Mathematics 2018

Universidad Complutense de Madrid, Spain.

GPA: 9.58/10. Dissertation: *The Ambrose–Singer Theorem*. Advisor: Prof. Marco Castrillón López.

B.Sc. in Mathematics 2017

University of Málaga, Spain.

GPA: 9.17/10.

Research Interests

Differential Geometry; Geometric Analysis; Homogeneous Geometry; Complex Hyperbolic Geometry; Invariant Connections; Conformal Geometry; Weyl Structures; Submanifold Geometry.

Publications

Accepted

[Car26b] José Luis Carmona Jiménez. “On Weyl Structures Reducible in the Direction of the Lee Form”. *Proceedings of the American Mathematical Society* **154**:(10) (2026). DOI: [10.1090/proc/17755](https://doi.org/10.1090/proc/17755).

[CC25] José Luis Carmona Jiménez and Marco Castrillón López. “Canonical Reductive Decomposition of Extrinsic Homogeneous Submanifolds”. *Note Mat.* **45**:(Supl. 1) (2025), pp. 117–128. DOI: [10.1285/i15900932v45s1p117](https://doi.org/10.1285/i15900932v45s1p117).

- [CCD25] José Luis Carmona Jiménez, Marco Castrillón López, and José Carlos Díaz-Ramos. “The Ambrose–Singer Theorem for Cohomogeneity One Riemannian Manifolds”. *Transformation Groups* (2025). DOI: [10.1007/s00031-025-09927-x](https://doi.org/10.1007/s00031-025-09927-x).
- [CC22a] José Luis Carmona Jiménez and Marco Castrillón López. “The Ambrose–Singer Theorem for General Homogeneous Manifolds with Applications to Symplectic Geometry”. *Mediterranean Journal of Mathematics* **19**:(6) (Nov. 2022), p. 280. DOI: [10.1007/s00009-022-02197-x](https://doi.org/10.1007/s00009-022-02197-x).
- [CC22b] José Luis Carmona Jiménez and Marco Castrillón López. “The homogeneous holonomies of complex hyperbolic space”. *Annals of Global Analysis and Geometry* **62**:(2) (July 2022), pp. 391–411. DOI: [10.1007/s10455-022-09852-2](https://doi.org/10.1007/s10455-022-09852-2).
- [CC20] José Luis Carmona Jiménez and Marco Castrillón López. “Reduction of Homogeneous pseudo-Kähler Structures by One-Dimensional Fibers”. *Axioms* **9**:(3) (2020). DOI: [10.3390/axioms9030094](https://doi.org/10.3390/axioms9030094).

Preprints

- [Car26a] José Luis Carmona Jiménez. *On generalized imaginary Spin^c -Killing spinors*. arXiv. 2026. URL: <https://arxiv.org/abs/2605.07813>.
- [CGS26] José Luis Carmona Jiménez, Alejandro Gil-García, and C. S. Shahbazi. *Kenmotsu manifolds and spin-c Killing spinors*. arXiv. 2026.

Manuscripts in Preparation

- [CCL26] José Luis Carmona Jiménez, Marco Castrillón López, and Eugenia Loiudice. *The Ambrose–Singer Theorem for non-Regular Cohomogeneity One Riemannian Manifolds*. In preparation. 2026.

Teaching Experience

University Teaching

Riemannian Geometry	2024–2025
University of Bucharest, Romania	
Exercise sessions for Master’s students, 6 hours.	
Linear Algebra	2020–2022
Universidad Complutense de Madrid, Spain	
Exercise sessions for first-year undergraduate students, 75 hours.	
Mathematical Methods	2021–2022
Universidad Complutense de Madrid, Spain	
Exercise sessions for first-year undergraduate students, 15 hours.	
The Ambrose–Singer Theorem	2021
University of Santiago de Compostela, Spain	
Invited research seminar course, 10 hours.	

Secondary Education

Mathematics Teacher	2022–2024
I.E.S. Mar de Alborán and I.E.S. Las Lagunas, Spain	
Full-time teaching in secondary and upper-secondary education. Approximately 1296 teaching hours.	

Long Research Stays, including Forthcoming Stays

Mathematisches Forschungsinstitut Oberwolfach	10/01/2027–02/04/2027
Oberwolfach, Germany.	
Forthcoming twelve-week postdoctoral research stay as an Oberwolfach Leibniz Fellow.	
Riemann Center for Geometry and Physics, Leibniz Universität Hannover	01/10/2026–31/12/2026

Hannover, Germany.

Forthcoming three-month postdoctoral research stay at the Institute of Differential Geometry.
Hosts: Prof. Roger Bielawski and Prof. Knut Smoczyk. Administrative formalization pending.

Philipps-Universität Marburg

15/09/2025–30/11/2025

Marburg, Germany.

Postdoctoral research stay with Dr. Eugenia Loiudice at the Faculty of Mathematics and Informatics.

Philipps-Universität Marburg

01/05/2022–31/07/2022

Marburg, Germany.

Predocutorial research stay with Prof. Dr. habil. Ilka Agricola at the Faculty of Mathematics and Informatics.

University of Santiago de Compostela

20/09/2021–20/12/2021

Santiago de Compostela, Spain.

Predocutorial research stay with Prof. José Carlos Díaz Ramos at the Department of Geometry and Topology.

Short Research Stays

Philipps-Universität Marburg

23/06/2025–27/06/2025

Marburg, Germany.

Research stay with Dr. Eugenia Loiudice at the Faculty of Mathematics and Informatics.

University of Málaga

17/10/2024–28/10/2024

Málaga, Spain.

Short research stay with Prof. Marián Cañadas at the Department of Geometry and Topology.
Talk: *On the history of Ambrose–Singer Theorems*.

Research Projects

Geometric Structures on Manifolds, Symmetry and Field Theory

01/09/2025–

31/08/2027

Role: Participant.

Principal Investigator: Prof. Marco Castrillón López. Funding entity: Ministry of Science and Innovation, Spain. Funding amount: 91,250.00 EUR. Reference: PID2024-156578NB-I00.

Conformal Aspects of Geometry and Dynamics

01/01/2024–30/06/2026

Role: Participant.

Principal Investigator: Prof. Andrei Moroianu. Funding entity: NextGenerationEU, through the Planul Național de Redresare și Reziliență, Romania. Funding amount: 6.000.000 RON, approximately 1.200.000 EUR. Reference: PNRR-III-C9-2023-I8 grant CF 149/31.07.2023.

Geometrical, Topological and Combinatorial Structures Associated with Dynamical Systems

01/09/2022–31/08/2025

Role: Participant.

Principal Investigators: Prof. Marco Castrillón López and Prof. Luis Giraldo. Funding entity: Ministry of Science and Innovation, Spain. Funding amount: 108,900.00 EUR. Reference: PID2021-126124NB-I00.

Combinatorics and Dynamics: Geometric and Topological Aspects in Manifolds

01/01/2019–31/12/2021

Role: Participant.

Principal Investigators: Prof. Marco Castrillón López and Prof. Luis Giraldo. Funding entity: Ministry of Science and Innovation, Spain. Funding amount: 68,002.00 EUR. Reference: PGC2018-098321-B-I00.

Algebra and Geometry: Differential Geometry, Symplectic Geometry and Geometric Mechanics

2020

Role: Participant during my research contract at ICMAT.

Reference: 20205-CEX001.

Competitive Fellowships and Research Contracts

Oberwolfach Leibniz Fellow

10/01/2027–02/04/2027

Mathematisches Forschungsinstitut Oberwolfach (MFO), Oberwolfach, Germany.

Postdoctoral fellowship awarded by the Mathematisches Forschungsinstitut Oberwolfach / Leibniz Association for a forthcoming 12 weeks research stay.

Riemann Fellowship

01/10/2026–31/12/2026

Riemann Center for Geometry and Physics, Leibniz Universität Hannover, Germany.

Postdoctoral research fellowship for a forthcoming three months research stay at the Institute of Differential Geometry. Hosts: Prof. Roger Bielawski and Prof. Knut Smoczyk. Administrative formalization pending.

International Fellows@FCMH Research Fellowship

15/09/2025–30/11/2025

Philipps-Universität Marburg, Germany.

Postdoctoral mobility fellowship funded by the Forschungscampus Mittelhessen for a research stay with Dr. Eugenia Loiudice at the Faculty of Mathematics and Informatics. Research topic: cohomogeneity-one actions and related problems in Differential Geometry.

Postdoctoral Research Contract

16/09/2024–30/06/2026

Simion Stoilow Institute of Mathematics of the Romanian Academy (IMAR), Bucharest, Romania.

Postdoctoral contract associated with the project *Conformal Aspects of Geometry and Dynamics*. Funding: PNRR-III-C9-2023-I8 grant CF 149/31.07.2023, NextGenerationEU/PNRR, Romania; IMAR, Romanian Academy.

FPI Predoctoral Fellowship

01/10/2020–31/08/2022

Universidad Complutense de Madrid, Spain.

Competitive predoctoral research fellowship/contract for the training of research personnel, funded by the Spanish Ministry of Science, Innovation and Universities. Department of Algebra, Geometry and Topology, Faculty of Mathematical Sciences.

Research Contract

01/07/2020–30/09/2020

Instituto de Ciencias Matemáticas (ICMAT), Madrid, Spain.

Predocctoral research contract funded by the Spanish National Research Council (CSIC) through the Severo Ochoa Centre of Excellence extraordinary grant 20205-CEX001. Research activity in Differential Geometry, Symplectic Geometry and Geometric Mechanics.

Research Fellowship for Departmental Collaboration

06/11/2017–06/07/2018

Universidad Complutense de Madrid, Spain.

Fellowship for research collaboration in the Department of Algebra, Geometry and Topology. Advisor: Prof. Marco Castrillón López.

Organization of Scientific Events

Coorganizer of the Workshop: Women in Mathematics Day

12/05/2026

Simion Stoilow Institute of Mathematics of the Romanian Academy, Bucharest, Romania

<https://sites.google.com/view/wmd-imar/home>

Coorganizer of the Conference: Low-dimensional Complex and Conformal Geometry

01/06/2026–03/06/2026

Simion Stoilow Institute of Mathematics of the Romanian Academy, Bucharest, Romania

<https://sites.google.com/view/pnrrworkshopjune2026/main>

Conference Talks and Posters

Recent Trends in Differential Geometry 5-11 July 2025

Banach Conference Center in Bedlewo, Poland

Contributed talk: *Every Kenmotsu manifold is the real Hyperbolic space.*

RSME's 7th Congress of Young Researchers January 2025

Bilbao, Spain.

Contributed talk: *Symplectic invariant connections.*

The 2nd International Conference on Differential Geometry – ICDG Fez 2024 October 2024

Fez, Morocco.

Contributed talk: *Cohomogeneity one invariant connections.*

Conference Differential Geometry and its Applications August 2022

Hradec Králové, Czech Republic.

Contributed talk: *The Ambrose–Singer Theorem for symplectic geometry.*

Symmetry and Shape II October 2021

University of Santiago de Compostela, Spain.

Contributed talk: *On reductive locally homogeneous Fedosov manifolds.*

Online Summer School on Geometry and Topology August 2021

University of Hradec Králové.

Contributed talk: *Fedosov Homogeneous Structures.*

Third BYMAT Conference: Bringing Young Mathematicians Together December 2020

Online.

Contributed talk: *The Ambrose–Singer Theorem for general homogeneous manifolds with applications to symplectic geometry.*

Symmetry and Shape I October 2019

University of Santiago de Compostela, Spain.

Poster: *Characterization of simply-connected homogeneous spaces with geometric structures.*

Conference Differential Geometry and its Applications September 2019

Hradec Králové, Czech Republic.

Poster: *A generalization of the Ambrose–Singer Theorem.*

13th ICMAT International Summer School on Geometry, Mechanics and Control July 2019

ICMAT, Madrid, Spain.

Contributed talk: *Locally homogeneous Riemannian spaces.*

Conference Attendance

Conformal Geometry and More – A Conference in Honor of Liviu Ornea July 2025

Bucharest, Romania.

Attendee.

MAM III: Quaternion-Kähler Manifolds & the LeBrun–Salamon Conjecture February 2025

Philipps-Universität Marburg, Germany.

Attendee.

Second BYMAT Conference: Bringing Young Mathematicians Together May 2019

ICMAT, Madrid, Spain.

Attendee.

12th ICMAT International Summer School on Geometry, Mechanics and Control
 July 2018
 University of Santiago de Compostela, Spain.
 Attendee.

Research Seminars and Courses

Seminar Courses

Seminar Course: The Ambrose–Singer Theorem 20/09/2021–20/12/2021
 University of Santiago de Compostela, Spain.
 Invited ten-hour research seminar course at the Department of Mathematics.

Research Seminars

Geometry Seminar 24/06/2025
 Philipps-Universität Marburg, Germany.
 Talk: *The Ambrose–Singer Theorem for Cohomogeneity One Riemannian Manifolds.*

Geometry Seminar 09/06/2025
 IMAR, Bucharest, Romania.
 Talk: *On Weyl structures reducible in the direction of the Lee form.*

Geometry Seminar 11/04/2025
 IMAR, Bucharest, Romania.
 Talk: *The Ambrose–Singer Theorem for Cohomogeneity One Manifolds.*

Geometry and Topology Seminar 21/10/2024
 University of Málaga, Spain.
 Talk: *On the history of Ambrose–Singer Theorems.*

Geometry Seminar 15/10/2024
 IMAR, Bucharest, Romania.
 Talk: *Ambrose–Singer Theorems III.*

Geometry Seminar 08/10/2024
 IMAR, Bucharest, Romania.
 Talk: *Ambrose–Singer Theorems II.*

Geometry Seminar 01/10/2024
 IMAR, Bucharest, Romania.
 Talk: *Ambrose–Singer Theorems.*

Geometry and Topology Seminar 04/04/2022
 University of Málaga, Spain.
 Talk: *The Ambrose–Singer Theorem.*

Outreach

VII Ph.D.ay Mathematics 03/07/2023
 Universidad Complutense de Madrid, Spain.
 Outreach talk: *Tesellaciones y espacios homogéneos.* English title: *Tessellations and homogeneous spaces.*

IV Conference on Career Opportunities in Mathematics 12/04/2023
 University of Málaga, Spain.
 Round table participant on academic and professional career paths in Mathematics.

Divulgation Seminar 17/11/2021
 University of Santiago de Compostela, Spain.

Outreach talk: *Teselaciones y espacios homogéneos*. English title: *Tessellations and homogeneous spaces*.

V Ph.D.ay Mathematics

24/06/2021

Universidad Complutense de Madrid, Spain.

Outreach talk: *Las geometrías homogéneas del espacio hiperbólico complejo*. English title: *The Homogeneous Geometries of Complex Hyperbolic Space*.

III Ph.D.ay Mathematics

02/07/2019

Universidad Complutense de Madrid, Spain.

Outreach talk: *El teorema de Ambrose–Singer. Un puente entre el Álgebra, la Geometría y el Análisis*. English title: *The Ambrose–Singer Theorem: A Bridge between Algebra, Geometry, and Analysis*.

Languages

Spanish: native. English: advanced.

Technical Skills

- **Tools:** LaTeX, Git/GitHub, HTML.